

FANUC Robot **Small Hand Guidance**

OPERATOR'S MANUAL

MAROBESHGD02191E REV. A

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

1

Small Hand guidance

The small hand guidance function enables the robot to be moved by operating the hand guidance unit. The robot moves to the direction of the joystick's motion. The connection cable is not necessary because the small hand guidance unit communicates with the robot controller (iPendant) wirelessly

Handling

This function can be used to pick up and carry workpiece and ease these operations.(Fig.1(a))

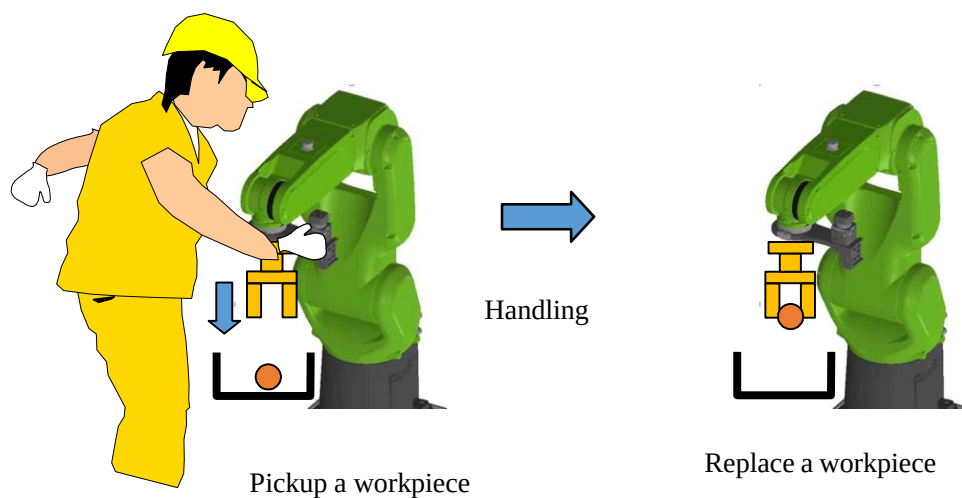


Fig. 1(a) Hand guidance operation



CAUTION

- In the hand guidance operation, if you feel danger, for example, there is a risk that the work piece collides yourself or the stand etc., please release the hand guidance unit immediately.
- If hand guidance function does not resume after collision, move the robot in opposite direction by jog operation.

Direct Teach

It is possible to do Direct Teaching with a Hand Guidance unit. It can automatically generate TP program. There are following two methods.

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EDIT	DATE	DESIG.	DESCRIPTION	FANUC CORPORATION	2/44

There are two types of teaching methods.

1. Recording full path

While the “Teach” button is pushed in the hand guidance operation, the TCP positions are recorded continuously. If the “Teach” button is pushed again, the teaching mode is exited and the robot program is automatically generated. (Fig.1(b))

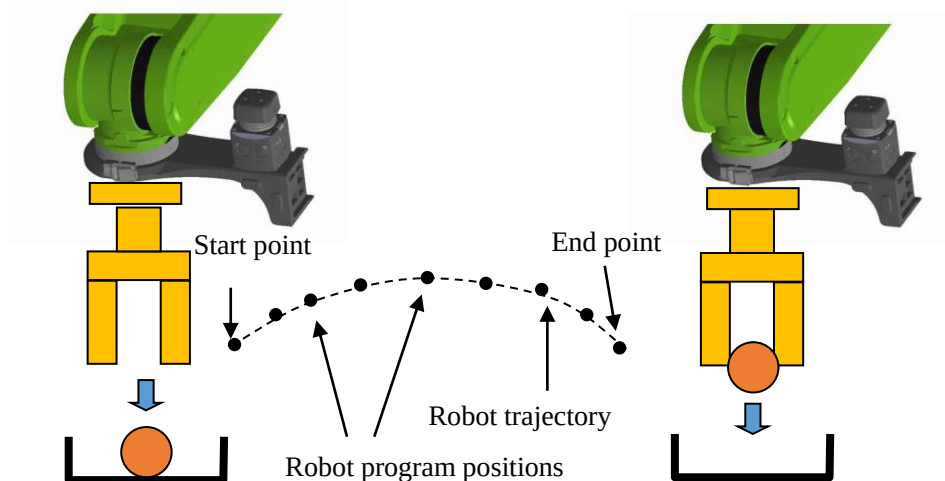


Fig. 1(b) Recording full path

2. Recording specified points

Once the “Teach” switch is pushed in the hand guidance operation, one point is recorded. If the “Teach” switch is pushed at multiple times, multiple points are added. After ending the teaching mode, a robot program is generated by using all recorded points automatically. (Fig.1(c))

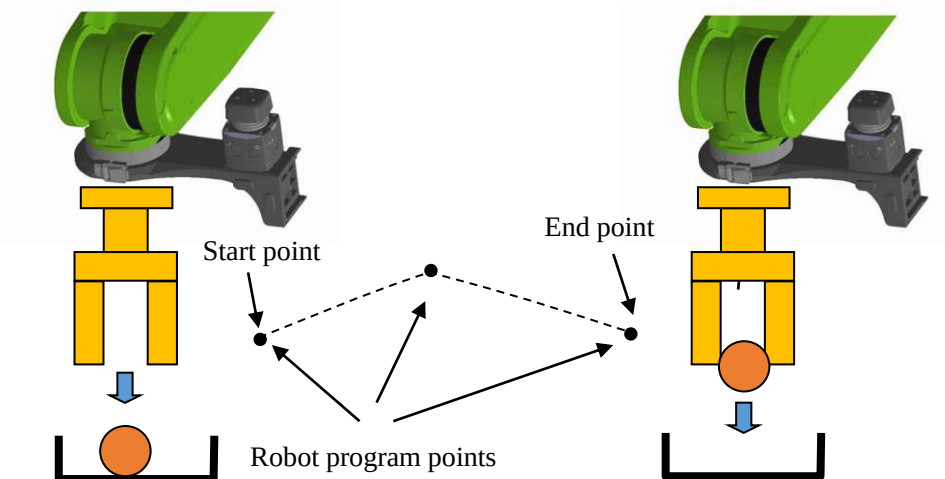


Fig. 1(c) Recording specified points

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1.1 System configuration

1.1.1 Hardware

Following hardware are necessary to use small hand guidance on top of robot and a controller.

- Small hand guidance unit A05B-1425-K601 (part number : A97L-0318-0670)
- Adapter for Small hand guidance A05B-1425-K611 (only for CR-4iA, CR-7iA, CR-7iA/L)
- Memory : SRAM 2MB or 3MB

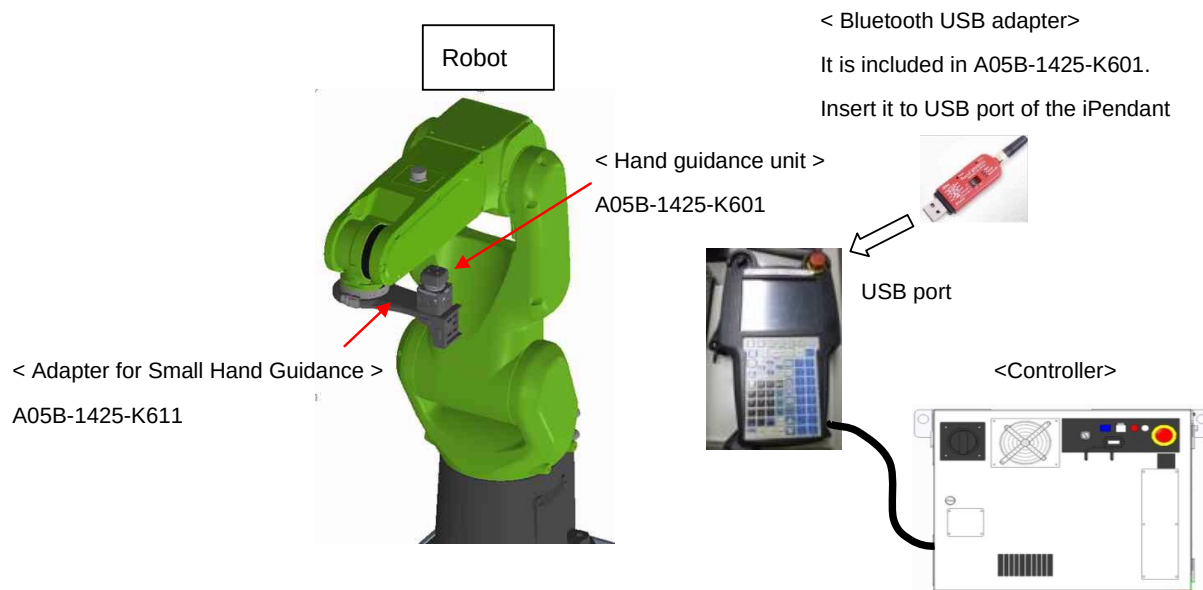


Fig. 1.1.1 System layout

1.1.2 Software

The software option A05B-2600-S505 (Small Hand Guidance) is necessary for the small hand guidance operation.

The A05B-2600-S505 can't be ordered simultaneously with following software options.

- Hand Guidance A05B-2600-J841
- Force Control Basic A05B-2600-J876
- 3-axis Force sensor A05B-2600-J844
- Line Tracking A05B-2600-J512

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1.2 Supported model

As of Jan. 2019

- Supported Robot models are Collaborative robot CR-4iA, CR-7iA CR-7iA/L, CR-15iA, CR-35iA.

1.3 Wireless and Safety Certificate

The small hand guidance unit communicates with the controller wirelessly (Bluetooth).

As of Jan, 2019, it has authentication for wireless device in Japan, North America (USA and Canada), EU (see following list), EFTA (see following list), Turkey and Vietnam.

It can not be used in other countries, yet.

- EU: (28 countries)

Estonia, Latvia, Lithuania, Cyprus, Malta, Belgium, Netherlands, Luxembourg, Germany, France, Italy, UK, Ireland, Denmark, Greece, Spain, Portugal, Austria, Finland, Sweden, Poland, Hungary, Czech Republic, Slovenia, Slovakia, Romania, Bulgaria, Croatia

- EFTA: (4 countries)

Iceland, Norway, Liechtenstein, Switzerland

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2

Small Hand guidance unit

2.1

Outline of Hand Guidance Unit

The movable part of the stick has 4 degrees of freedom.[X,Y,Z and around Z]

- In Translation mode, the robot moves in X/Y/Z (3 directions).
- In Rotation mode, the robot moves around X/around Y/around Z (3 directions).
- Translation mode and Rotation mode is switched by pressing 'Mode' button.

3. Translation in Z

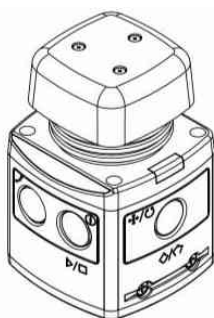
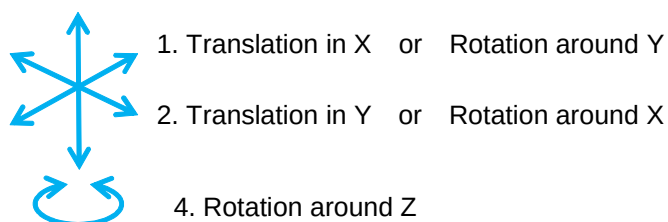


Fig. 2.1(a) Joystick

The unit has four buttons just as shown in Fig.2.1(b) There are LEDs (color : Yellowish green) above each buttons to show the status of them.

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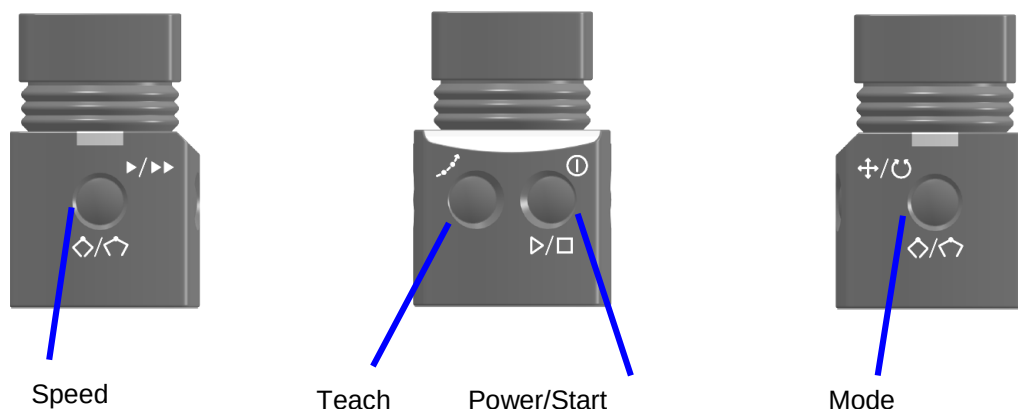


Fig. 2.1(b) Button

The meaning of each button and LED is explained in Table 2.1.

Button	Pictograph	Function	LED display status
Power / Start		Power ON / OFF Press and hold for more than 3 seconds to turn on the power Press and hold for more than 3 seconds again to turn off the power	Power ON: lit Power OFF: Off
		After power on, Hand guide start / end	Handguide start: blink (every 1 second) Handguide end: lit
Teach		Recording full path mode (*), start / end of recording Recording specified points mode (*) Recording of taught points	In case of Recording full path mode It lights up when it starts up, and goes out when it finishes In case of Recording specified points mode Lit only while button is pressed
Mode		Translation / Rotation	In case of Translation (X,Y,Z) : Lights off In case of Rotation (around X, around Y, around Z):lit
Speed		High speed / Low speed	High speed :Lights off Low speed: lit

*) Refer to section “3.2.5.1 How to do direct teaching”

If Mode and Speed buttons are pressed simultaneously, the gripper can be opened or closed.

Button	Pictograph	Function	LED display status
Mode + Speed		Gripper open / Gripper close Toggle gripper open / close if released at the same time	The mode LED and the speed LED blink (300 ms interval) while pressed simultaneously If released, they return to the state before simultaneous pushing.

Table 2.1 Button and LED

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2.2 External dimensions

The External dimensions of the small hand guidance unit and the Bluetooth USB adapter are shown in Fig.2.2(a) and Fig.2.2(b).

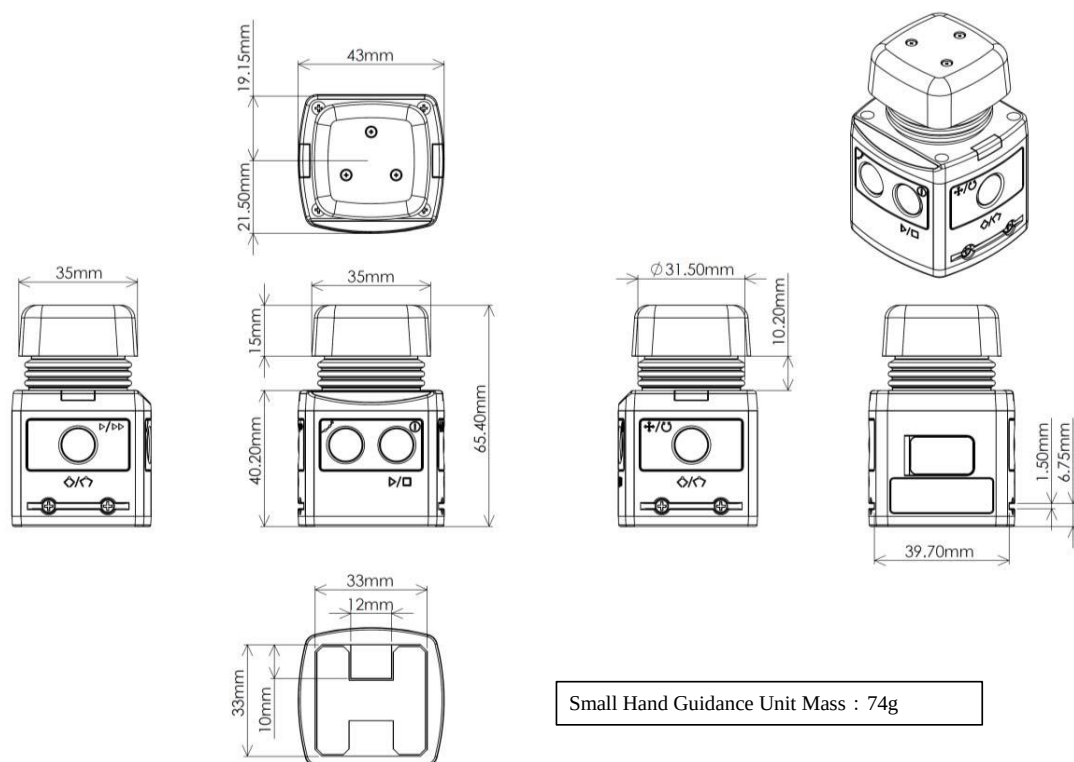


Fig.2.2(a) The External dimension of the small hand guidance unit

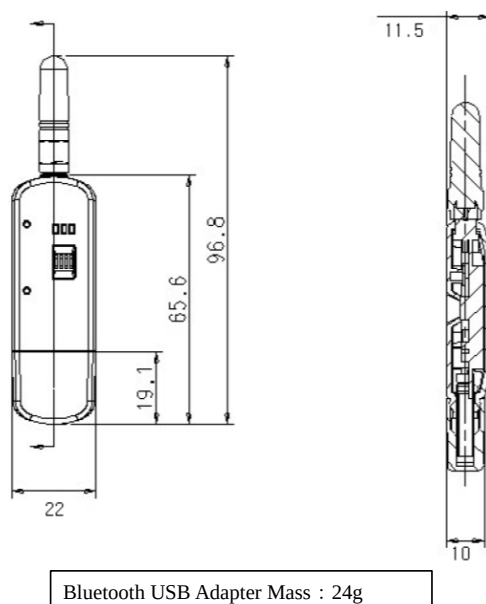


Fig.2.2(b) The External dimension of the Bluetooth USB adapter

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2.3 Notation of label

The position and contents of label of the small hand guidance unit and the Bluetooth USB adapter are shown in Fig.2.3(a) and Fig.2.3(b).

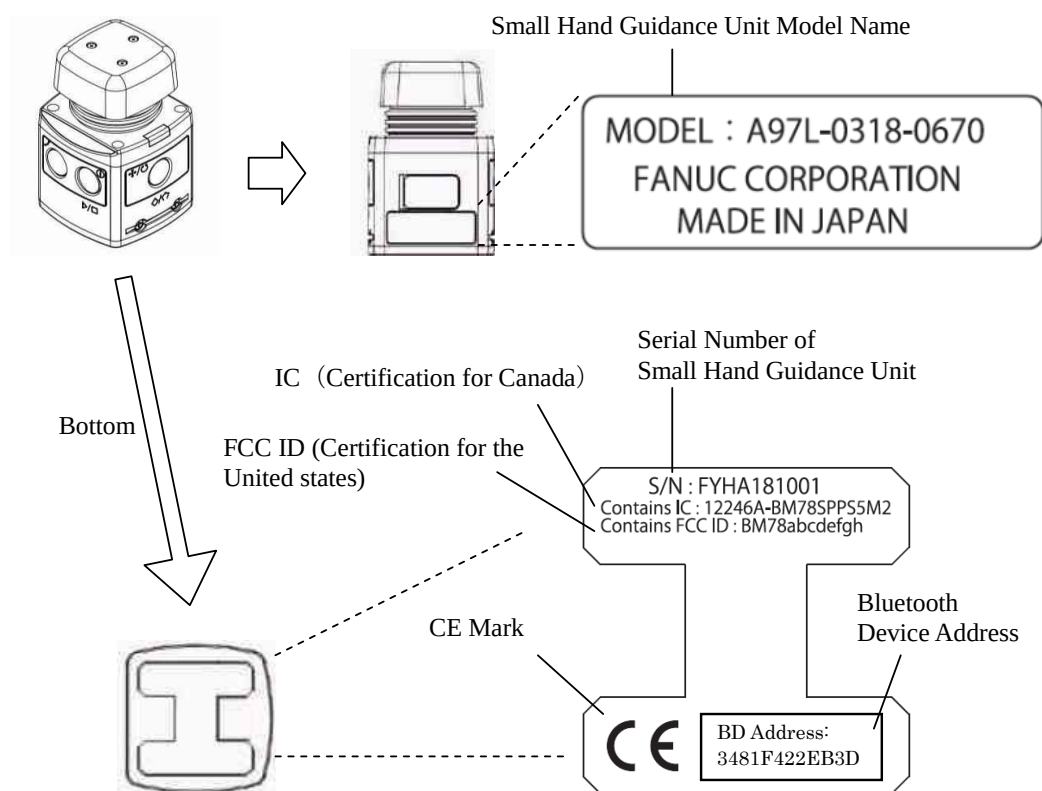


Fig.2.3(a) The position and contents of label of the small hand guidance unit

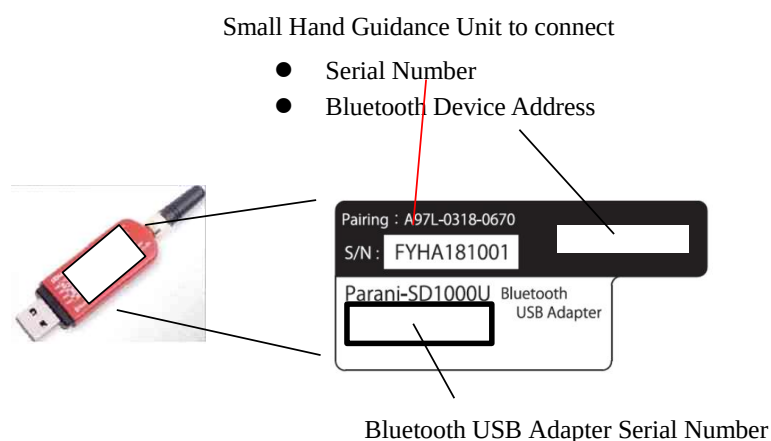


Fig.2.3(b) The position and contents of label of the Bluetooth USB adapter

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2.4 Charging the battery

- Micro USB port is on the side without buttons. Charge the battery from here
- It works for 8 hours with full charge
- The cable for charging has to be prepared by the user
Ex.: Micro USB (Type B male) - USB 2.0 Type A for smartphones
- Please do not execute hand guidance while charging

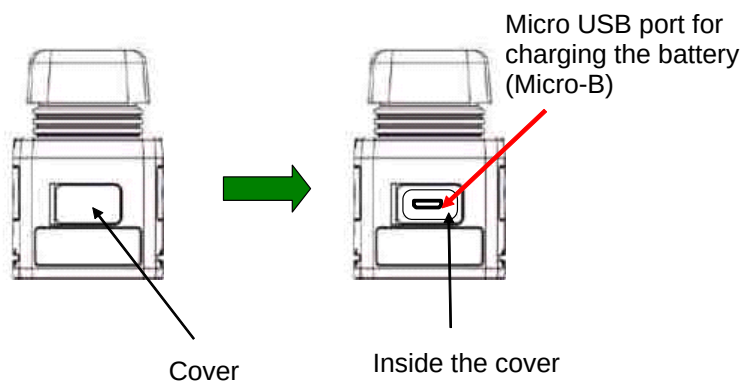


Fig.2.4(a) Micro USB port for charging battery

While charging battery, following two LEDs are lit. They will be turned off automatically after charging is completed.

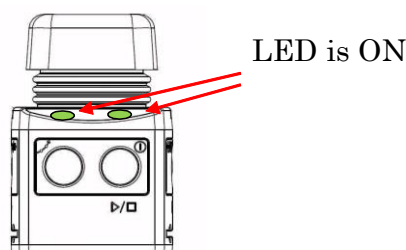


Fig.2.4(b) LEDs are ON while charging

2.5 Motion range

When hand guidance is used, the motion range of each robot axis is narrower than standard range by 5 deg. For example,

- In case of CR-4iA, the standard J5 axis range is between -100 deg and 100 deg. The range is between -95 deg and 95 deg while hand guidance is used.
- In case of CR-7iA, CR-7iA/L, the standard J5 axis range is between -120 deg and 120 deg. The range is between -110 deg and 110 deg while hand guidance is used.

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2.6 Mounting the hand guidance unit onto the small collaborative robot

- 1 Mount the hand guidance unit onto the adapter (A05B-1425-K611).
There are grooves on both sides of the bottom of the unit. Align these grooves with the slide section on the adapter, and push in until you hear a click.

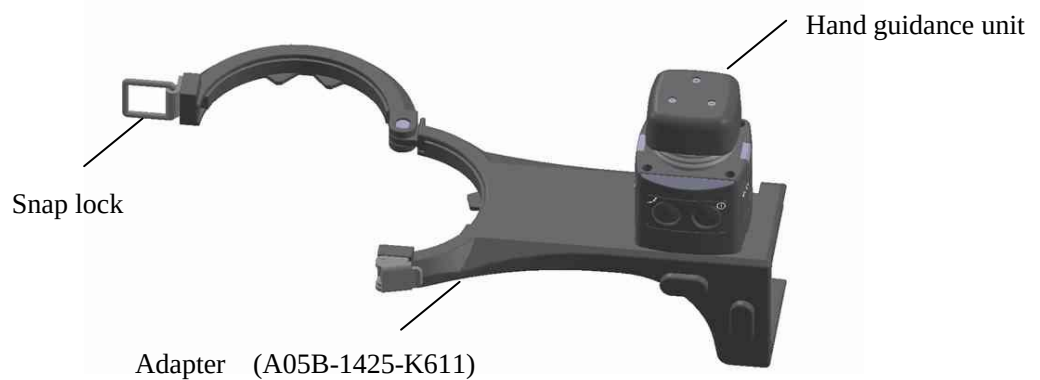


Fig. 2.6(a) Hand guidance unit and adapter

- 2 Set the robot's orientation to $(J2, J3, J4, J5) = (0, 0, 0, 0)$.

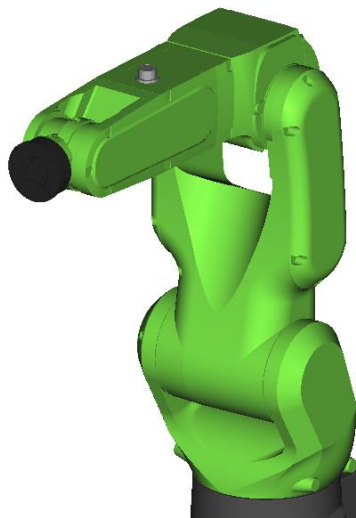


Fig. 2.6(b) Mounting orientation

- 3 Mount the adapter around the robot's mechanical flange. While keeping the top of the adapter flush with the surface of the mechanical flange as much as possible, ensure that the adapter's tabs are situated between the bolts of the flange, as shown in the figure. Finally, firmly close the snap lock.

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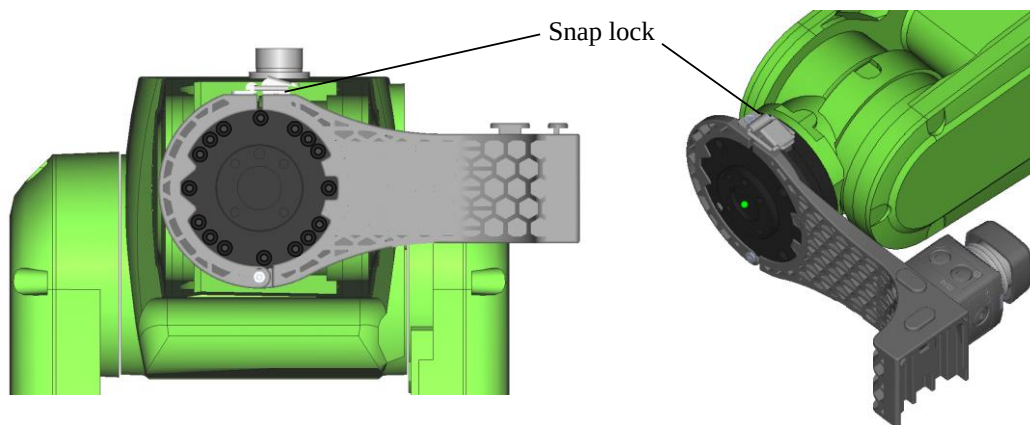


Fig. 2.6(c) Adapter mounting position

⚠ CAUTION

Do not use the J6 reducer bolt cover (A05B-1142-J001). If this part is used, the adapter cannot be mounted.

2.7 Robot operation during hand guidance

As shown in section 2.6, because the adapter is mounted around the robot flange, the hand guidance unit is situated above the robot's J6 axis.

- Even in translation mode, the robot does not move along the fixed Cartesian coordinate system. Rather, it moves in a gentle curve, as shown in fig. 2.7.
- In rotation mode, it rotates centered on the currently selected tool frame's origin (TCP).

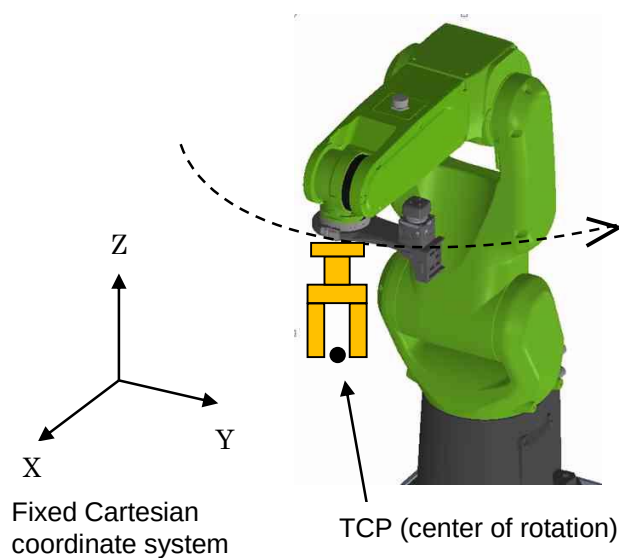


Fig. 2.7 Image of robot operation direction

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2.8 Notes

- The hand guidance unit communicates with the controller wirelessly. The communication may be unstable if there are Wi-Fi spots (2.4GHz band) or strong radio transmission source nearby.
- Be sure to use the same combination of hand guidance unit and Bluetooth USB adapter that was in the same box at the time of delivery. Wireless communication fails with wrong combination. The serial number of the hand guidance unit is written on a label of the Bluetooth USB adapter. (See Fig.2.3(b))
- Do not disassemble, repair nor remodel the hand guidance unit.
- Do not exert excessive force to joystick
- Built-in battery

Lithium polymer secondary battery is built in.

Not applicable to aviation dangerous goods and export controlled goods, but pay attention to transportation method.

- Can be transported it as it is at delivery packaging
- When transporting the hand guidance units taken out from the delivery box in another box, do not exceed 2 pieces in the same packing
- When transporting three or more in the same packing, the following indication is required on the surface of the box

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① Lithium battery mark

② Describing the necessary wording on the air waybill



Contact :

Source : IATA Air Dangerous Goods Regulations 58 th edition (2017)

Lithium ion batteries in
compliance with
Section II of PI967

- The mark is a rectangle hatched with diagonal lines.
- The minimum dimension of the mark is 120 mm (width) × 110 mm (height), the minimum width of hatching is 5 mm
- Hatching is red.
- Display UN3481 (United nation number)

Do not exchange battery. If it is necessary to exchange it, contact FANUC.

Warning

- There is a danger of explosion if the battery is replaced with a wrong type. Recycle or dispose of the replaced battery as an industrial waste, according to the laws and other rules in the country where the controller is installed and those established by the municipality and other organizations that have jurisdiction over the area where the controller is installed.
- Charge the battery at least once in 6 months even if the hand guidance unit is not used. It may not be able to be charged if the battery voltage is too low.

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3 Software for hand guidance

3.1 How to use

- 1 Connect the Bluetooth USB adapter to the USB port of the teach pendant (iPendant), as shown in fig. 3.1(a).



Fig. 3.1(a) Connection of Bluetooth USB adapter

- 2 Press the power button of the hand guidance unit for at least 3 seconds to turn on the power, as shown in fig. 3.1(b). Then, the hand guidance unit will automatically connect wirelessly to the Bluetooth USB adapter. When the connection is complete, the “Connect” LED of the Bluetooth USB adapter blinks. Be sure to use the hand guidance unit and Bluetooth USB adapter that were in the same box when they were delivered. A connection cannot be established with different combinations.

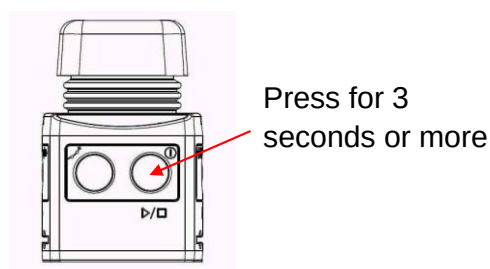


Fig. 3.1(b) Power on

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- 3 When “Connection State” shows “CONNECTED” on the hand guidance’s initial setting screen (refer to “3.2.1 Initial Setting”), wireless communication between the robot controller and the hand guidance unit will begin automatically.

During wireless communication, the following will happen:

- The “Serial-Tx/Rx” LED on the Bluetooth USB adapter blinks.
- The LED above the start button of the hand guidance unit lights continuously. (Fig. 3.1(c))
- “Operation State” shows “READY” on the initial setting screen.

If an alarm has occurred in the robot, “Operation State” shows “STOPPED” on the initial setting screen, and all the LEDs on the unit blink (Fig. 3.1(d)).

First, clear the cause of the robot alarm and reset it. After that, the operation state will automatically change to “READY”.

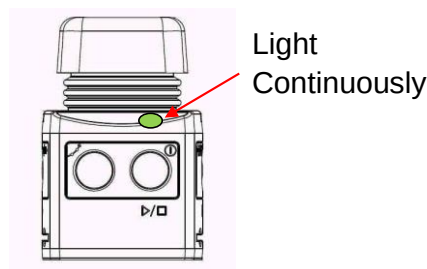


Fig. 3.1(c) “Operation State” is “READY”

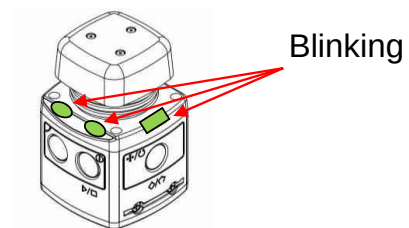


Fig. 3.1(d) “Operation State” is “STOPPED”

- 4 Press the start button on the unit. The unit’s start button LED will blink, and hand guidance operation will begin. (Fig. 3.1(e))

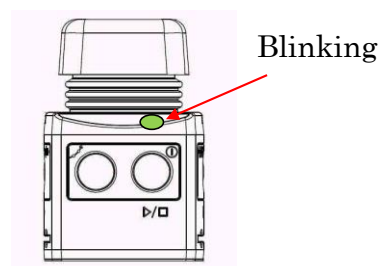


Fig. 3.1(e) Hand guidance operating

- 5 Move the joystick to operate the robot.
- 6 Press the start button on the unit again. The unit’s start button LED will light continuously, and hand guidance operation will stop.
- 7 At this point, you can turn off the unit’s power or remove the Bluetooth USB adapter.

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3.2 Hand Guidance Utility Screen

Select [MENU] → [1 UTILITIES] → [Hand Guidance] to open the “UTILITIES Hand guidance” screen. In this screen, parameter setting for hand guidance can be set.

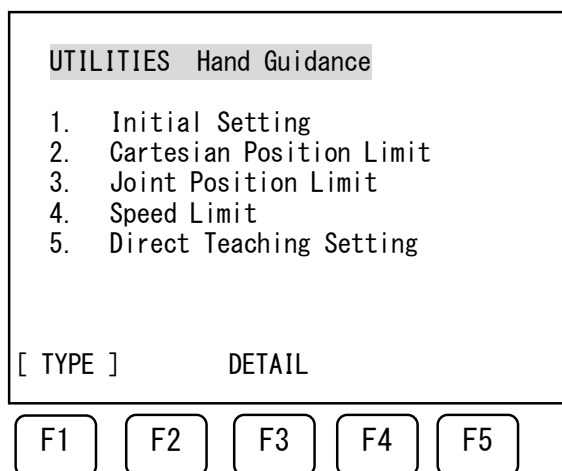


Fig. 3.2 Hand guidance utility screen

Move the cursor to each item to select it and open each detailed screen as follows.

Screen	Detail
Initial Setting	In this screen, the wireless connection status of the small hand guidance are shown. The unit mounting and the gripper open/close program can be configured.
Cartesian Position Limit	In this screen, the limitation of the Cartesian position of the robot TCP during the hand guidance can be set. Set the interfering area of the robot in order to prevent the collision.
Joint Position Limit	In this screen, the limitation of the joint position during the hand guidance can be set.
Speed Limit	In this screen, Cartesian and rotation speed limit of the rotation center of the handle and the TCP for DCS can be set.
Direct Teaching Setting	Direct teaching function is to record the robot position during the hand guidance and then generate robot program automatically.

3.2.1 Initial Setting

In this screen, the wireless connection status of the small hand guidance are shown. The unit mounting and the gripper open/close program can be configured.

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Initial Setting
1/1

Unit Status

1 Connection Status : DISCONNECT

2 Operation Status : NOT_READY

3 Battery Remain : ----

Unit Mounting Setting

4 Mounting Method : DEFAULT

Default Mounting

(Move Robot to Position(J2~J5=0))

5 Attachment Orient : RIGHT

6 Mount to Attachment Side : DISABLE

7 Power SW Orient : UPWARD

Custom Mounting

(Unit Center Position)

PowerSW Ornt:+X, SpeedSW Ornt:+Y)

8 X: 0.0 Y: 0.0 Z: 0.0

9 W: 0.0 P: 0.0 R: 0.0

Gripper Setting

10 Gripper Open/Close SW : DISABLE

11 Switch Status : ---

12 Open Program : [*****]

13 Close Program : [*****]

TP Prog Exec after Hand Guidance

14 Resume Original Path : ENABLE

EXIT
RESET

F1

F2

F3

F4

F5

Fig. 3.2.1(a) Initial setting screen

Parameters

Unit Status

- Connection State

This item shows the wireless connection state between the hand guidance unit and the controller.

- **CONNECTED:** The hand guidance unit and the controller are connected. To establish a connection, the following two conditions must be met:

- ✧ The Bluetooth USB adapter is inserted in the USB port of iPendant.
- ✧ The power of the small hand guidance unit is turned on.

When Connection State is “CONNECTED”, wireless communication will be started automatically.

- **DISCONNECT:** The hand guidance unit and the controller are not connected. One of the following may be the reason they are disconnected.
 - ✧ The Bluetooth USB adapter is not inserted in the USB port of iPendant.
 - ✧ The power of the small hand guidance unit is not turned on.
 - ✧ A nearby device, such as a Wi-Fi hotspot, may be emitting strong radio signals that are interfering with the Bluetooth communication.

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Default : DISCONNECT

- Operation State

This item shows the operation state of the small hand guidance.

- READY: Hand guidance operations are possible. Hand guidance operations are possible when all of the following conditions are met:
 - ✧ Wireless communication is established.
 - ✧ No alarm has occurred in the robot.
- RUNNING: Hand guidance is operating.
- NOT READY: Wireless communication has not been established and hand guidance cannot be started.
- STOPPED: An alarm has occurred in the robot. First, clear the cause of the robot alarm, and then reset it.

Default : NOT READY

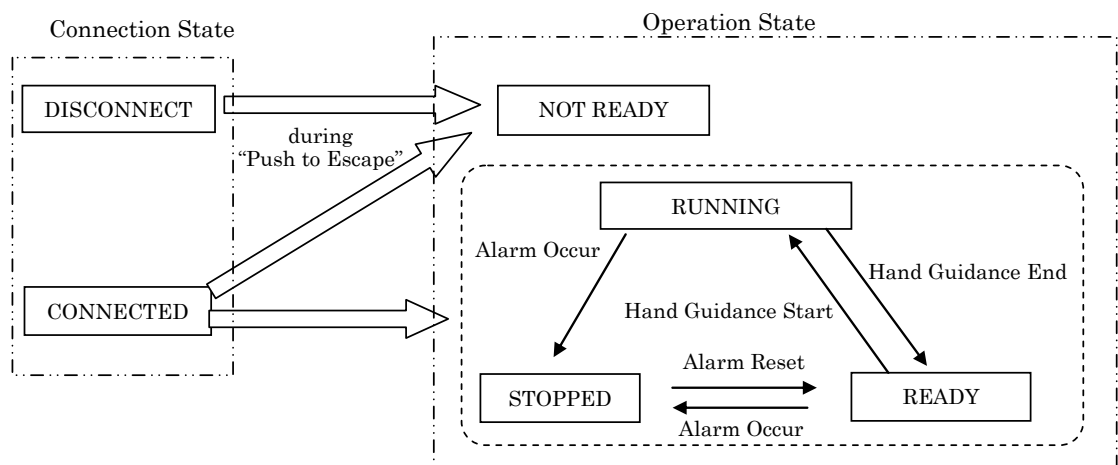


Fig. 3.2.1(b) Relationship between Connection State and Operation State

- Battery Remain

This item shows the amount of power remaining in the built-in battery of the hand guidance unit.

- ---: Remaining battery amount unknown.
- HIGH: The amount of remaining battery is high.
- MIDDLE: The amount of remaining battery is middle.
- LOW: The amount of remaining battery is low, and the battery needs to be recharged soon.
- ZERO: The amount of remaining battery is almost zero. Recharge the battery right away.
- CHARGING: The battery is charging.

Default : ---

Unit Mounting Setting

- Mounting Method: DEFAULT/CUSTOM

- For small collaborative robots, the mounting method can be set to DEFAULT or CUSTOM.

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- For robots other than small collaborative robots, the mounting method is set to CUSTOM automatically and cannot be changed.

- Default Mounting
- Move robot to Position(J2~J5=0))

Move the robot to position J2~J5=0, and then configure the following settings.

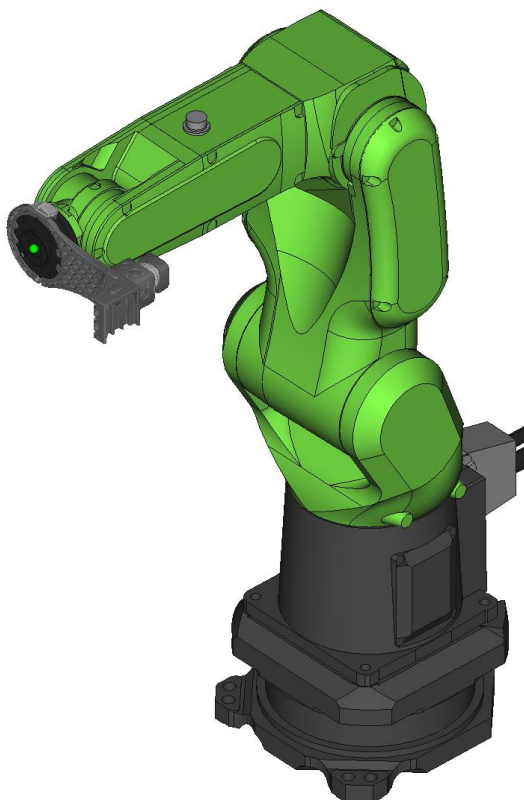


Fig. 3.2.1(c) Robot orientation for unit mounting

- Attachment Orient

This is the orientation of the attachment (adapter) when the operator is facing the robot.

- RIGHT: The attachment is oriented to the right.
- LEFT: The attachment is oriented to the left.

Default : RIGHT

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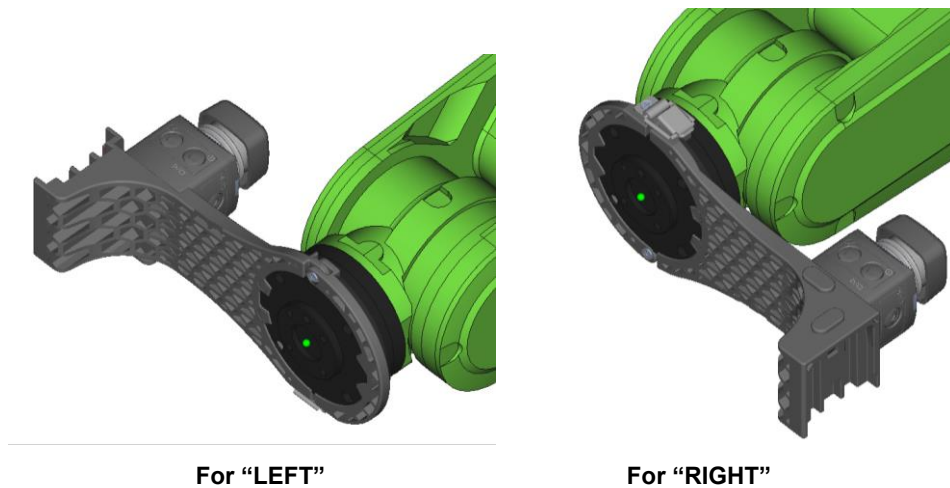


Fig. 3.2.1(d) Attachment orientation

- **Mount to Attachment Side**

Specify whether the unit is mounted on the side of the attachment (adapter).

- **DISABLE:** Mounted above the attachment.
- **ENABLE:** Mounted on the side of the attachment.

Default : DISABLE

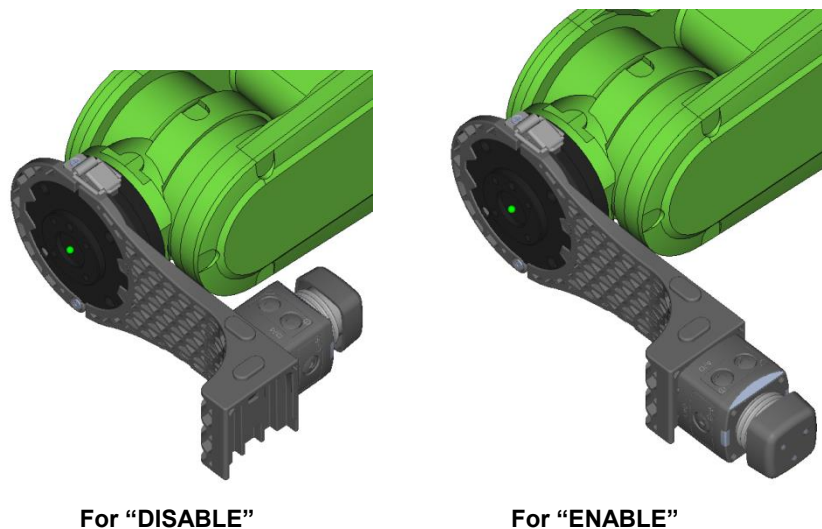


Fig. 3.2.1(e) Mounting on the side of the attachment

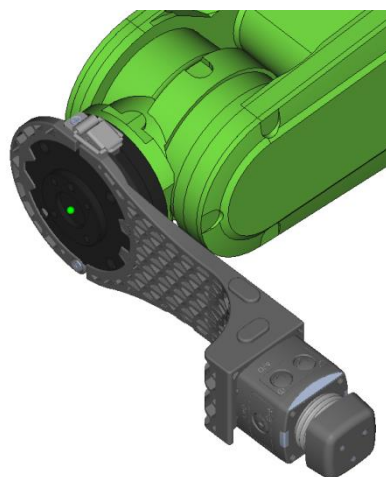
- **Power SW Orient**

Specify the orientation of the power switch.

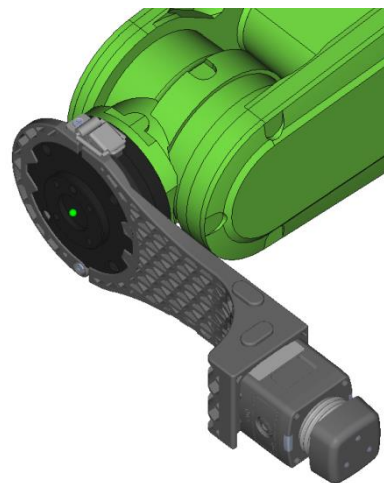
- **UPWARD:** The power switch is facing upward.
- **DOWNWARD:** The power switch is facing downward.

Default : UPWARD

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For “UPWARD”



For “DOWNWARD”

Fig. 3.2.1(f) Power switch orientation

- Custom Mounting
- (Unit Center Position, PowerSW Ornt:+X, SpeedSW Ornt:+Y)
- X: 0.0, Y: 0.0, Z: 0.0 [unit : mm]
- W: 0.0, P: 0.0, R: 0.0 [unit : deg]

Mount the unit to the hand of the robot and set the unit frame.

- The “Unit Center Position” is the center position of the top surface of the unit in the flange frame. It is the origin of the unit frame.
- The “PowerSW Ornt” is the direction of the power switch from the origin. It is +X direction of the unit frame.
- The “SpeedSW Ornt” is the direction of the speed switch from the origin. It is +Y direction of the unit frame.

Set X, Y and Z values of the top surface of the unit in the flange frame.

Set W, P and R values of the unit direction in the flange frame.

In the example of Fig. 3.2.1(g), the unit frame is = (200,0,-20,0,0,0)。

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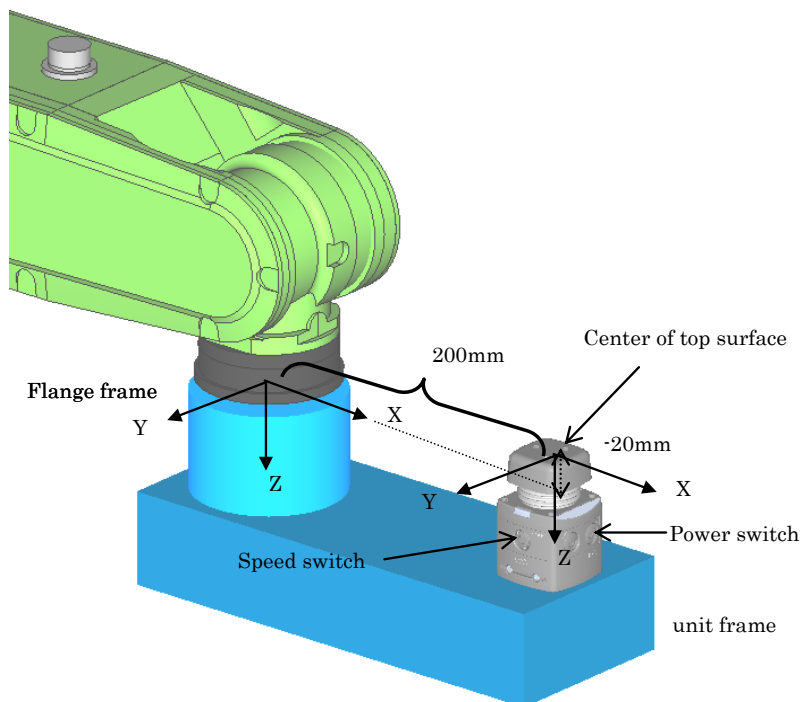


Fig. 3.2.1(g) Custom Mounting setting

Gripper Setting

- Gripper Open/Close SW :

To enable or disable the gripper open/close button.

- Enable: Enable the gripper open/close button.
- Disable: Disable the gripper open/close button.

Default : Disable

- Switch Status

This item indicates the current state of the Gripper Open/Close SW.

- ---: Uninitialized.
- OFF: The gripper is closed.
- ON: The gripper is open.

Default : ---

⚠ CAUTION

When the gripper open/close button is pushed or released, the gripped workpiece may drop down. A risk assessment for the whole robot system that the drop down of the workpiece is necessary.

- Open Program:

Set the name of the program that is executed when the gripper open/close button is switched from OFF to ON.

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Default : “*****”

- Close Program :

Set the name of the program that is executed when the gripper open/close button is switched from ON to OFF.

Default : “*****”

NOTE

If the payload number is changed in this open/close program, the motion group mask must be enabled.

The motion command must NOT be included into the open/close program.

TP Prog Exec after Hand Guidance

- Resume Original Path:

This parameter specifies if original path resume motion is done after TP program is paused and robot is moved by hand guidance. In other words, it specifies whether the robot returns to the paused position before resuming the program after hand guidance or not.

For detail about original path resume, please refer to “9.13 Original Path Resume” in “R-30iB/R-30iB Mate /R-30iB Plus CONTROLLER OPERATOR’S MANUAL (B-83284EN)”

- ENABLE: The robot returns to the paused position before resuming the program after hand guidance. But the robot does not return to the paused position and move to next point if the original path resume function is disabled.
- DISABLE: The robot does not return to the paused position and move to next point when resuming the program.

Default: ENABLE

Function Key

- F2 [RESET] : All parameters will be reset to the default value.

3.2.2 Cartesian Position Limit

3.2.2.1 List Screen of the Cartesian Position Limit

In this screen, a list of the Cartesian position limit is displayed. In the hand guidance operation, the status of all Cartesian position limit will be shown in this screen.

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Cartesian Pos. Limit List				
No.	ENB/DSBL	Area	Status	Comment
1	Disable	Inside	----	[]
2	Disable	Inside	----	[]
3	Disable	Inside	----	[]
4	Disable	Inside	----	[]
5	Disable	Inside	----	[]
6	Disable	Inside	----	[]
7	Disable	Inside	----	[]
8	Disable	Inside	----	[]
9	Disable	Inside	----	[]
10	Disable	Inside	----	[]

EXIT DETAIL

F1 F2 F3 F4 F5

Fig. 3.2.2.1 Cartesian Position Limit List Screen

3.2.2.2 Detail Screen of the Cartesian Position Limit

Select a line in the list screen of the Cartesian position limit and then pushing F3 “DETAIL” to display the details of the selected number of the limit condition as follows. In this screen, operating limit of the TCP on the Cartesian position during the hand guidance can be set.

Cartesian Pos. Limit Detail			
No.	[1]	Status:	[-----]
1	Comment	[	]
2	Enable/Disable :	Disable	
3	I/O Signal :	D0[0] =	---
4	User Frame No. :	0	
5	Tool Frame No. :	1	
6	Motion Area :	Inside	
7	Margin :	50.0 mm	
Position Limit			
	Current	Lower	Upper
8	X: 200.0	-3000.0	3000.0 mm
9	Y: 1.3	-3000.0	3000.0 mm
10	Z: -20.0	-3000.0	3000.0 mm

EXIT RESET

F1 F2 F3 F4 F5

Fig. 3.2.2.2(a) Cartesian Position Limit Detail Screen

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Parameters

- No. :

The number of the Cartesian position limit.

Its value is from 1 to 10.

- Status:

It is the state of the robot position in current Cartesian position limit. It has four types of states.

- Normal: The robot is in the normal area. (Hand guidance at normal speed is available.)
- Margin: The robot is in the margin area. (Hand guidance at slow speed is available.)
 - ✧ The Cartesian speed is slower than 20mm/s and the speed of the DCS TCP is slower than the speed limit of the DCS TCP.
 - ✧ The rotation speed is slower than 2deg/s.
- Over: The robot is out of the margin and normal area. (Can only move to return to the margin area.)
- -----: If the position limit is disabled or the specified I/O is OFF.

Default value : “-----”

- Comment:

Comment of this Cartesian position limit.

Default value: “*****”

- Enable/Disable:

To set whether to enable or disable the Cartesian position limit.

- Enable: The Cartesian position limit is always enabled.
- Disable: The Cartesian position limit is always disabled.
- By I/O: Enable/disable the Cartesian position limit according to the value of specified I/O.

Default value: Disable

- I/O Signal:

Specify the type (DI/DO/RI/RO/F(Flag)) and number of the I/O to be used to enable or disable the position limit. The value of the specified I/O is also shown.

- ✧ If the specified signal is ON, then the position limit is enabled.
- ✧ If the specified signal is OFF, then the position limit is disabled.

Default value : DI [0] = ---

- User frame number:

Specify the user frame number for current position and current position limit.

Default : 0

- Tool frame number:

Specify the tool frame number for current position and current position limit.

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Default : 1

- Motion area:

Specify the motion area to be inside or outside of the position limit.

Besides, there are following limits

- At most one position limit with the “Inside” motion area can be enabled at one time. If two or more position limits with the “Inside” motion area are enabled at one time, alarms are posted and robot stops.
- At most one position limit with the “Outside” motion area can be enabled at one time. If two or more position limits with the “Outside” motion area are enabled at one time, alarms are posted and robot stops.

Default: Inside

- Margin:

Set the margin of the Cartesian position limit.

Default: 50 mm

- Position limit

- Current

It shows the current TCP position in current user frame.

Default: 0

- Lower

- Upper

Set the lower and upper limit for current TCP position

- In the Cartesian position limit, the normal area and margin area exist.
- If current TCP position is in the normal area of the Cartesian position limit, the hand guidance can be operated normally.
- If current TCP position is in the margin area of the Cartesian position limit, the hand guidance can be operated in a slow speed.
- If current TCP position is not in the normal area nor the margin area of the Cartesian position limit, the hand guidance can be operated in a slow speed and can only move to the return to the margin area.
- In position limits with the “Inside” motion area, minimum width of X,Y,Z direction of limited area is 20mm.
- In position limits with the “Outside” motion area, minimum width of X,Y,Z direction of limited area is 20mm.

Default: Lower: -3000mm,

Upper: 3000mm.

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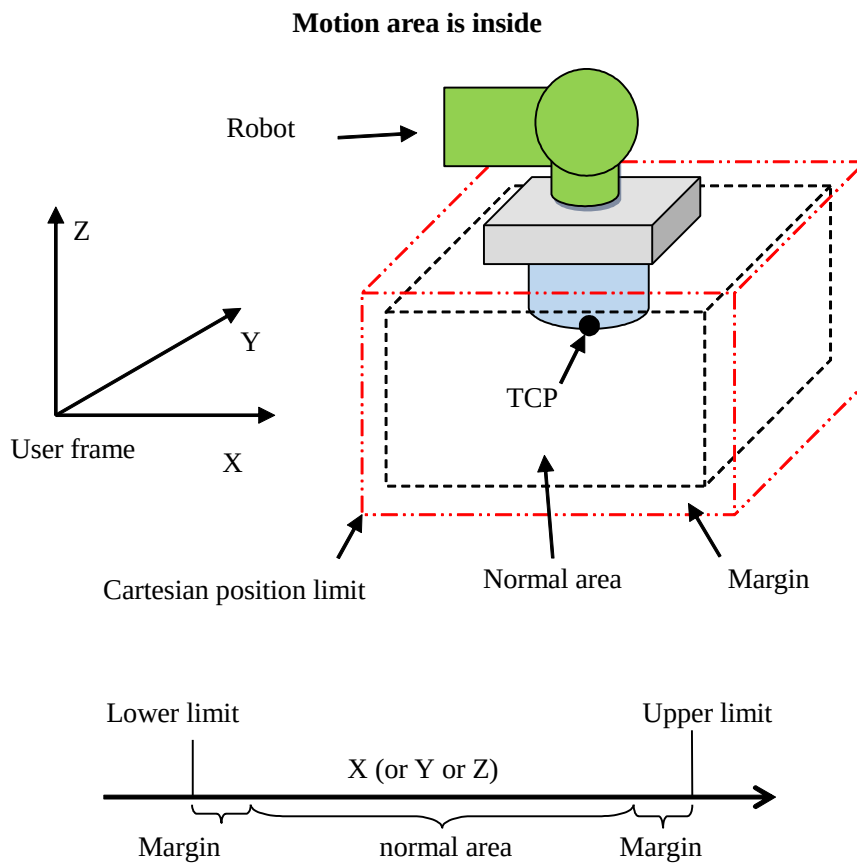


Fig. 3.2.2.2(b) Cartesian Position Limit with inside motion area

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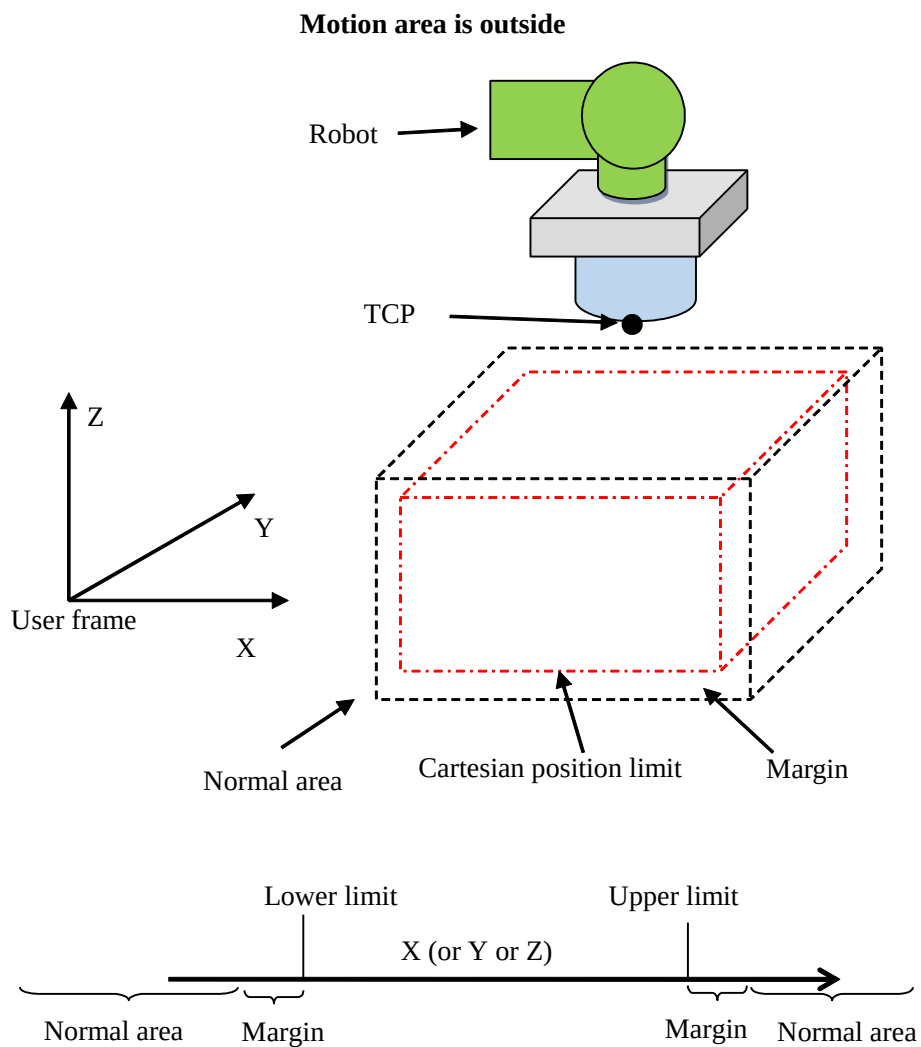


Fig. 3.2.2.2(c) Cartesian Position Limit with outside motion area

Function Key

- F2 [RESET] : All parameters will be reset to the default value.

3.2.3 Joint position limit

In this screen, set the joint position limit for the hand guidance.

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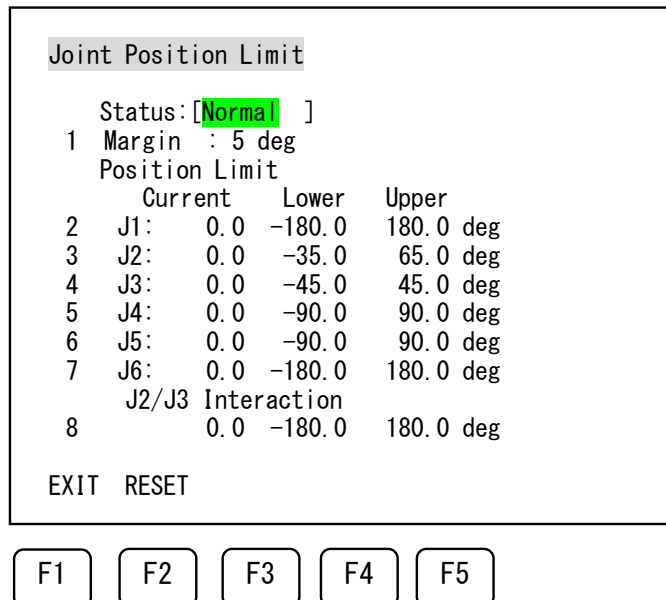


Fig. 3.2.3(a) Joint Position Limit screen

Parameters

● Status:

It is the state of all joint axes and J2/J3 interaction. It has three states.

- Normal: All joint axes and J2/J3 interaction are in the normal area. (Hand guidance at normal speed is available.)
- Margin(JNT): At least one joint axis or J2/J3 interaction is in the margin area and others are in the normal area. (Hand guidance at slow speed is available.)
 - ✧ The Cartesian speed is slower than 20mm/s and the speed of the DCS TCP is slower than the speed limit of the DCS TCP.
 - ✧ The rotation speed is slower than 2deg/s.
- Over: At least one joint axis or J2/J3 interaction is not in the normal area nor the margin area. (Hand guidance cannot be started.)
- MARGIN(SNGL): A joint axis is in the margin area near the robot's singular point. Operate in joint motion.
- SINGULAR: A joint axis is near the robot's singular point. Operate in joint motion.

Default: Normal

● Margin :

The joint position limit margin of each joint axis and J2/J3 interaction.

Default : 5 deg

● Position limit

● Current

It is the current position of each joint axis and J2/J3 interaction.

Default: 0

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- Lower
- Upper

Set the lower and upper position limit of each joint axis and J2/J3 interaction.

- In the joint position limit, the normal area and margin area exist.
- If current joint position is in the normal area, the hand guidance can be operated in normal speed mode.
- If current joint position is in the margin area, the hand guidance can be operated in slow speed mode.
- If current joint position is not in the normal area nor the margin area, the hand guidance can be operated in a slow speed and can only move to the return to the margin area.

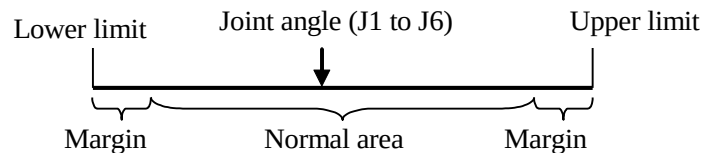


Fig. 3.2.3(b) Operational area

Default: Lower limit of J5 axis is -90deg, and upper limit of J5 axis is 90deg.

Limit of other axis are as follows.

Lower “System Axis Limits” +5deg,

Upper “System Axis Limits” -5deg

(The System Axis Limits screen can be displayed by selecting “Menu” -> “0—Next—” -> “System” -> “Axis limits”)

Lower limit of J2/J3 interaction is -57deg, and upper limit of J2/J3 interaction is 85deg.

Function Key

- F2 [RESET] : All parameters will be reset to the default value.

3.2.4 Speed limit

In this screen, Cartesian and rotation speed limit for the rotation center of the handle and the TCP of DCS can be set.

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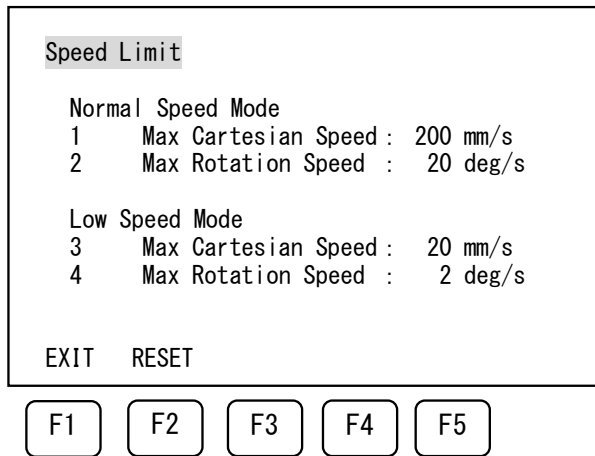


Fig. 3.2.4(a) Speed Limit screen

Parameters

- Normal speed mode
If “Speed” button on the handle is not being pushed and the robot is in the normal area, the hand guidance can be operated at the normal speed mode.
In this mode, the robot speed will not exceed the following speed limit.
- Max Cartesian speed
It is the max Cartesian speed of the rotation center of the handle and the TCP of DCS.
Default: 200 mm/s
- Max rotation speed
It is the max rotation speed of the rotation center of the handle.
Default: 20deg/s
- Slow speed mode
If “Speed” button on the handle is being pushed and the robot is in the normal area, the hand guidance will be operated at the slow speed mode.
In this mode, the robot speed will not exceed the following speed limit.
- Max Cartesian speed
It is the max Cartesian speed of the rotation center of the handle and the TCP of DCS.
Default: 20 mm/s
- Max rotation speed
It is the max rotation speed of the rotation center of the handle.
Default: 2deg/s

Function Key

- F2 [RESET] : All parameters will be reset to the default value.

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3.2.5 Direct teaching setting

Direct teaching function can record the robot positions in the hand guidance and generate robot program automatically. With this function, the complicated teaching operation can be simplified.

3.2.5.1 How to do direct teaching

There are two modes of position recording methods for the direct teaching:

- Recording full path.
- Recording specified points.

Recording full path

The full path during the hand guidance operation can be recorded continuously. There are following methods.

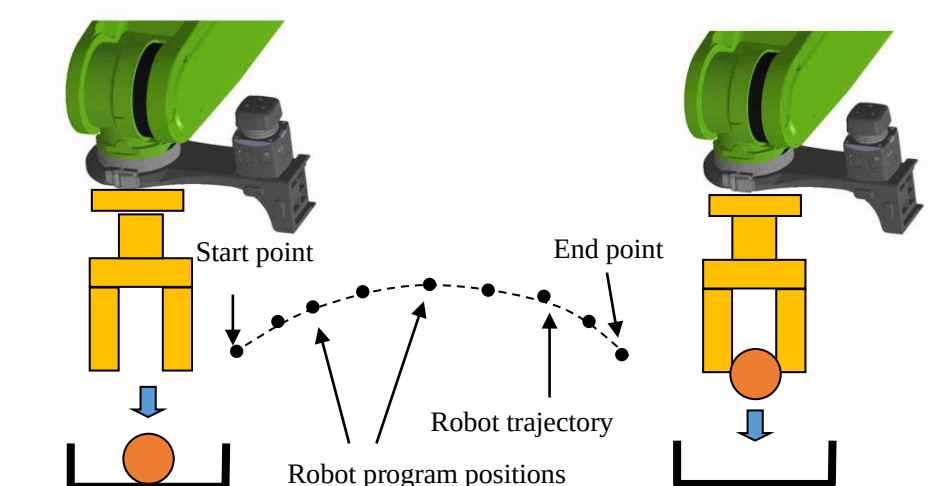


Fig. 3.2.5.1(a) Recording full path mode

Operation Method (1)

This is the method under the standard settings. Teaching is started and stopped with the teach button on the hand guidance unit (refer to table 2.1). During hand guidance operation, the robot's path is recorded and a TP program is automatically generated.

1. Press the hand guidance unit's teach button to start teaching. The LED on the teach button lights.

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2. Move the robot to the desired position and orientation. During this time, the robot's path is being recorded. Also, by enabling "Rec Gripper Open/Close" and pressing the mode button and the Speed button simultaneously, commands for opening or closing the gripper can be added.
3. To stop teaching, press the hand guidance unit's teach button again. The LED on the teach button turns off. The TP program is automatically generated with the name specified in "Program Name" below.

Operation Method (2)

This method is used to start and stop teaching according to the values of I/O signals instead of using the hand guidance unit's teach button. During hand guidance operation, the robot's path is recorded and a TP program is automatically generated.

1. On the direct teaching setting screen, set "Start/End Teach Method" to "By I/O Signal", and then specify the type and number of I/O to use in "By I/O Signal". Disable "Record Multiple Path".
2. Teaching starts when the signal above changes from OFF to ON.
3. Press the teach button, and the LED on the teach button lights.
4. Move the robot to the desired position and orientation. During this time, the robot's path is being recorded. Also, by enabling "Rec Gripper Open/Close" and pressing the mode button and the Speed button simultaneously, commands for opening or closing the gripper can be added.
5. Press the teach button, and the LED on the teach button turns off.
6. To record further movements, repeat steps 3 to 5.
7. Recording stops when the signal changes from ON to OFF. The TP program is automatically generated with the name specified in "Program Name".

Operation Method (3)

This method is used to start and stop teaching according to the values of I/O signals instead of using the hand guidance unit's teach button. During hand guidance operation, the robot's path is recorded and then divided into multiple sub TP programs that are created.

1. On the direct teaching setting screen, set "Start/End Teach Method" to "By I/O Signal", and then specify the type and number of I/O to use in "By I/O Signal". Enable "Record Multiple Path".
2. Teaching starts when the signal above changes from OFF to ON.
3. Press the teach button, and the LED on the teach button lights.
4. Move the robot to the desired position and orientation. During this time, the robot's path is being recorded. Also, by enabling "Rec Gripper Open/Close" and pressing the mode button and the Speed button simultaneously, commands for opening or closing the gripper can be added.
5. Press the teach button, and the LED on the teach button turns off. One subprogram is automatically generated. The subprogram name is [Program Name]_xx, where xx is a sequential number starting from 01.
6. To create further subprograms, repeat steps 2 to 5.
7. Recording stops when the signal changes from ON to OFF. The TP program is automatically generated with the name specified in "Program Name". The subprograms generated with the steps above are called from this TP program.

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Operation Methods (4) and (5)

In these methods, the starting and stopping of teaching is done manually. The operation steps are almost identical to those in option methods (2) and (3) above, but instead of the I/O signals in steps 2 and 7, teaching is started and stopped according to the value in “BY HAND” under “Start/End Teach Method” on the direct teaching setting screen.



CAUTION

If direct teach is executed again after the program is automatically generated, the program will be automatically overwritten. Before performing the next operation, save the program as necessary.

Record specified points

The robot’s position is recorded the instant that the hand guidance unit’s teach button is pressed.

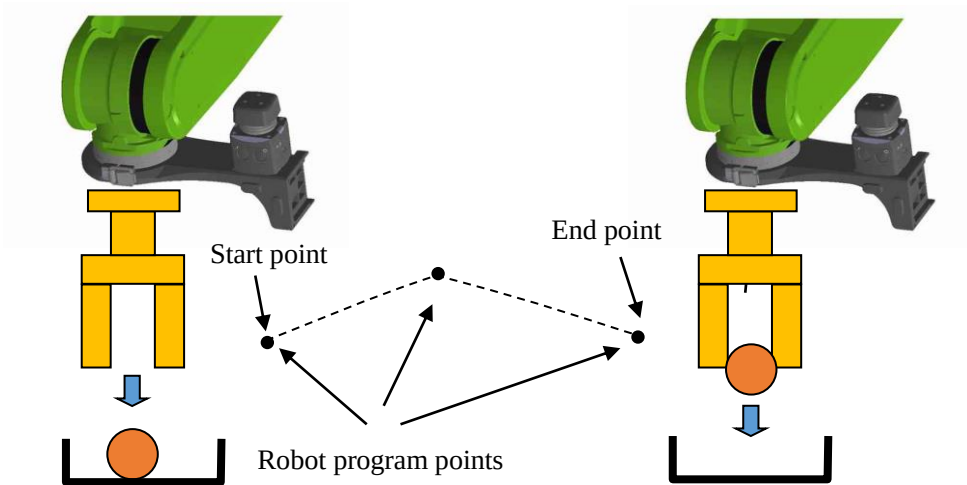


Fig. 3.2.5.1(b) Record specified points mode

The following two methods are available.

Operation Method (1)

In this method, teaching starts and stops according to the value of the I/O signal. The teach points during hand guidance operation are recorded and a TP program is automatically generated. “Record Multiple Path” cannot be enabled.

- 1. On the direct teaching setting screen, set “Start/End Teach Method” to “By I/O Signal”, and then specify the type and number of I/O to use in “By I/O Signal”. Also, set “Teaching Mode” to “POINT”.
- 2. Teaching starts when the signal above changes from OFF to ON.

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3. Move the robot to the desired position and orientation. Pressing the hand guidance unit's teach button (refer to table 2.1) at the desired position records that point. Pressing the teach button multiple times records multiple points. Also, by enabling "Rec Gripper Open/Close" and pressing the mode button and the Speed button simultaneously, you can add commands for opening or closing the gripper.
4. Recording stops when the signal changes from ON to OFF. The TP program is automatically generated with the name specified in "Program Name".

Operation Method (2)

In this method, the starting and stopping of teaching is done manually. The operation steps are almost identical to those in option method (1) above, but instead of the I/O signals in steps 2 and 4, teaching is started and stopped according to the value in "BY HAND" under "Start/End Teach Method" on the direct teaching setting screen.



CAUTION

If direct teaching is executed again after the program is automatically generated, the program will be automatically overwritten. Before performing the next operation, save the program as necessary.

3.2.5.2 Direct teaching setting screen

Select the "Direct Teaching Setting" in the "UTILITIES Hand Guidance" screen to open "Direct Teaching Setting" screen.

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Direct Teaching Setting

Teaching Setting

1. Start/End Teach Method : BY TEACH SW
2. By Hand : END
3. By I/O Signal : DO[0]
4. Teaching Mode : PATH
5. Record Multiple Path : DISABLE
6. Rec Gripper Open/Close : DISABLE
7. Rec Custom SW Output : DISABLE

TP Program Auto Generation

8. Program Name: HG_PROG_01
9. Cartesian Speed : 200 mm/s
10. Rotation Speed : 20 deg/s
11. Distance Threshold : 1.0 mm
12. Orient Change Thres. : 5.0 deg

EXIT RESET

F1

F2

F3

F4

F5

Fig. 3.2.5.2 Direct teaching setting screen

Parameters

Teaching Setting

- Start/End Teach Method

Select this parameter to set start or end teaching method, including BY TEACH SW/BY HAND/BY I/O.

Default: By teach SW

- When “BY TEACH SW” is selected for “Start/End Teach Method”

By pressing the unit’s teach switch, the LED for teaching lights, and position recording starts. By pressing the teach switch again, the LED for teaching turns off, position recording finishes, and the TP program is generated using the recorded position.

In this case, “Teaching Mode” below is set to “PATH” and cannot be changed. Also, “Record Multiple Path” is also set to “DISABLE” and cannot be changed.

- When “BY HAND” is selected for “Start/End Teach Method”

Select Start or End for “BY HAND” to start or end teaching.

➤ Start: Reset the direct teaching status and start direct teaching.

➤ End: End direct teaching and generate robot program with recorded positions.

Default: End

- When “BY I/O” is selected for “Start/End Teach Method”

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Teaching starts and ends using the specified I/O (DI/DO/RI/RO/F(Flag)) in “BY I/O Signal”, .

When the specified I/O changes from OFF to ON, teaching starts. When the specified I/O changes from ON to OFF, teaching ends, and the TP program is generated using the recorded position.

Default : DO[0]

- Teaching Mode

Set the mode of the direct teaching. If “Start/End Teach Method” is “BY TEACH SW”, Teaching Mode cannot be set to “POINT”.

- Path: To record the full path in the hand guidance.
- Point: To record specified points in the hand guidance.

Default: Path

- Record Multiple Path

Used to split into multiple paths and record. The maximum number of paths is 99. If the method for starting/ending teaching is “BY HAND” or “BY I/O”, this parameter can be set to ENABLE. If “Start/End Teach Method” is “BY TEACH SW”, this parameter cannot be set to ENABLE.

- DISABLE: Do not record multiple paths.
- ENABLE: Record multiple paths.

Default : DISABLE

- Rec Gripper Open/Close

Record the gripper open/close operation.

- DISABLE: Do not record the gripper open/close operation.
- ENABLE: Record the gripper open/close operation.

Default : DISABLE

- Rec Custom SW Output

This parameter is always “DISABLE” for the small hand guidance.

Default : DISABLE

TP Program Auto Generation

- Program Name

Specify the program name to be generated.

Default: ”HG_PROG_01”

- Cartesian Speed

Set the Cartesian speed of the generated robot program points. The range of values is 1~750 mm/s.

If the “Distance Threshold” is too small, the speed of the generated robot program points may be slower than this value.

Default: 200 mm/s

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- Rotation speed

The rotation speed of the generated robot program points. The range of values is 1~100 deg/s.
Default: 20 deg/s

- Distance Threshold

The maximum distance between the recorded points and the robot trajectory in the generated robot program. The range of values is 0.1~20 mm.

The smaller this value is, the more teaching points are set in a curve of an edge so that the path becomes closer to the actual robot trajectory.

Default: 1mm

- Orient Change Threshold.

The maximum orient change between the teach points of the generated robot program. The range of values is 1~20 deg.

The smaller this value is, the more teaching points are set when the orient changes so that the path becomes closer to the actual robot trajectory.

Default: 5deg

Function key

- F2 「RESET」 : All parameters will be reset to the default value.

3.2.5.3 Sample of generated TP program

When “Record Multiple Path” and “Rec Gripper Open/Close” are disabled

Only HG_PROG_01.TP like the one below is generated.

HG_PROG_01.TP

```

1  UFRAME_NUM=0
2  UTOOL_NUM=1
3  L P[1] 200mm/sec CNT100
4  L P[2] 200mm/sec CNT100
5  ...
6  L P[3] 200mm/sec CNT100

```

- If the Teaching Mode is PATH, the position type for the teach point is “CNT100”.
- If the Teaching Mode is POINT, the position type for the teach point is “FINE”.

When “Record Multiple Path” is disabled and “Rec Gripper Open/Close” is enabled

When HAND_CLOSE.TP was set for the gripper “Close Program” and HAND_OPEN.TP was set for the gripper “Open Program” on the initial setting screen.

Only HG_PROG_01.TP like the one below is generated.

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HG_PROG_01.TP

```

1  L P[1] 200mm/sec CNT100
2  L P[2] 200mm/sec FINE
3  CALL HAND_CLOSE ← Close the gripper
4  ...
5  L P[3] 200mm/sec CNT100
6  L P[4] 200mm/sec FINE
7  CALL HAND_OPEN ← Open the gripper

```

- If the Teaching Mode is PATH, the position type for the teach point just before the gripper open/close operation is always “FINE”, and “CNT100” for all other cases.
- If the Teaching Mode is POINT, the position type for the teach point is “FINE”.

When “Record Multiple Path” is enabled and “Rec Gripper Open/Close” is disabled

The main program HG_PROG_01.TP and subprogram HG_PROG_01_xx.TP (xx=01~99) like the ones below are generated.

HG_PROG_01.TP

```

1.  CALL HG_PROG_01_01
2.  CALL HG_PROG_01_02
...

```

HG_PROG_01_01.TP

```

1.  L P[1] 200mm/sec CNT100
2.  L P[1] 200mm/sec CNT100
3.  ...
4.  L P[2] 200mm/sec CNT100

```

When “Record Multiple Path” and “Rec Gripper Open/Close” are enabled

When HAND_CLOSE.TP was set for the gripper “Close Program” and HAND_OPEN.TP was set for the gripper “Open Program” on the initial setting screen.

The main program HG_PROG_01.TP and subprogram HG_PROG_01_xx.TP (xx=01~99) like the ones below are generated.

HG_PROG_01.TP

```

1.  CALL HG_PROG_01_01
2.  CALL HG_PROG_01_02
...

```

HG_PROG_01_01.TP

```

1  L P[1] 200mm/sec CNT100
2  L P[2] 200mm/sec FINE
3  CALL HAND_CLOSE
4  ...
5  L P[3] 200mm/sec CNT100
6  L P[4] 200mm/sec FINE
7  CALL HAND_OPEN

```

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3.3 Notice

- The nearby strong vibration source may worsen the performance of the hand guidance.
- The time limit of one hand guidance operation is two minutes. If the hand guidance operation exceeds the time limit, alarm happens and the robot stops.
- TP programs with motion group except Gripper Open /Close programs, cannot start execution in the hand guidance operation. If user attempts to start the TP program, robot stops with an alarm and hand guidance operation is suspended. Gripper Open/Close programs are specified in the “Initial Setting” screen (refer to subsection 3.3.1).
- When any TP program or jogging is executing, the program and the jogging is paused if the start switch on the unit is pushed and status of hand guidance changes to active.
- When contact stop happened during hand guidance, the robot stops and hand guidance is terminated.
- How to pass the area near the singularity point
 - If the mode button is turned OFF: (only Linear motion)

If the robot enters the margin area near the singularity points, the robot speed will be slowed down automatically. If the robot moves to pass the margin area to be nearer to the singularity point, robot will be stop automatically.
 - If the mode button is turned ON: (only Rotatory motion)

If the robot enters the margin area near the singularity points, the robot speed will be slowed down automatically and can pass the area near the singularity point at a non-translational motion. After leaving the area near the singularity point, the robot can moves in a normal mode.

3.4 Alarm code

3.4.1 Servo alarms

SRVO-485 Hand Guidance overspeed

Cause: Hand Guidance speed exceeds limit.

Remedy: Check Hand guidance setting.

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3.4.2 Force alarms

FORC-681 H/G Inside motion area error.

Cause: In the hand guidance operation, two or more Cartesian Position Limits with Inside motion area are enabled.

Remedy: Please enable only one Cartesian Position Limit with inside motion area in the hand guidance operation.

FORC-682 H/G Outside motion area error.

Cause: In the hand guidance operation, two or more Cartesian Position Limits with Outside motion area are enabled.

Remedy: Please enable only one Cartesian Position Limit with outside motion area in the hand guidance operation.

FORC-683 H/G Timeout error.

Cause: Hand guidance timeout error.

Remedy: Please finish a hand guidance operation within two minutes.

FORC-684 H/G Position Config changed.

Cause: During the direct teaching process, the position configuration has been changed.

Remedy: During the direct teaching process, please do not change the position configuration between Flip and Non-flip, or between Up and Down, or between Back and Front.

FORC-685 H/G initial setting error.

Cause: Hand guidance initial setting error.

Remedy: Please check the hand guidance initial setting screen.
To use Gripper Open/Close SW, please set as follows.
(a) Gripper Open/Close SW -> Enable.
(b) Open Program/Close Program -> Correct programs.

FORC-686 H/G I/O Signal setting Error.

Cause: Hand guidance I/O Signal setting error.

Remedy: Please check the hand guidance I/O Signal setting in the Cartesian Pos. Limit Detail screen.

FORC-687 H/G Too many recorded positions.

Cause: Too many recorded positions.

Remedy: Please reduce total recording time, or increase the Distance Threshold or Orient Change Threshold. and do the Direct Teaching again.

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FORC-688 H/G Too many teach points.**Cause:** Too many TP program teach points.**Remedy:** Please reduce total recording time, or increase the Distance Threshold or Orient Change Threshold. and do the Direct Teaching again.**FORC-689 H/G System is in error Status.****Cause:** System is in error status, therefore the hand guidance can not be started.**Remedy:** Please clear the error by pushing RESET button, and then try the hand guidance again.**FORC-690 H/G ENBL input is OFF.****Cause:** ENBL input is OFF, therefore the hand guidance can not be started.**Remedy:** Please set the ENBL input to ON and try the hand guidance again.**FORC-691 H/G HOLD input if ON.****Cause:** HOLD input is ON, therefore the hand guidance can not be started.**Remedy:** Please set the HOLD input to OFF and try the hand guidance again.**FORC-693 H/G Hand Guidance is running.****Cause:** H/G Hand Guidance is running.**Remedy:** Please end the hand guidance and then try again.**FORC-694 H/G Hand Guidance out of motion area.****Cause:** Robot is out of hand guidance motion area.**Remedy:** Jog the robot into the hand guidance motion area, and then try again.**FORC-695 H/G Hand Guidance stopped.****Cause:** Hand guidance motion has been stopped.**Remedy:** Please check the robot status, then try again.**FORC-701 SHG battery low.****Cause:** Small hand guidance unit's battery remaining is low.**Remedy:** Please recharge the unit as soon as possible.

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FORC-702 SHG battery zero.

Cause: Small hand guidance unit's battery remaining is almost zero.

Remedy: Please recharge the unit before using it.

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